

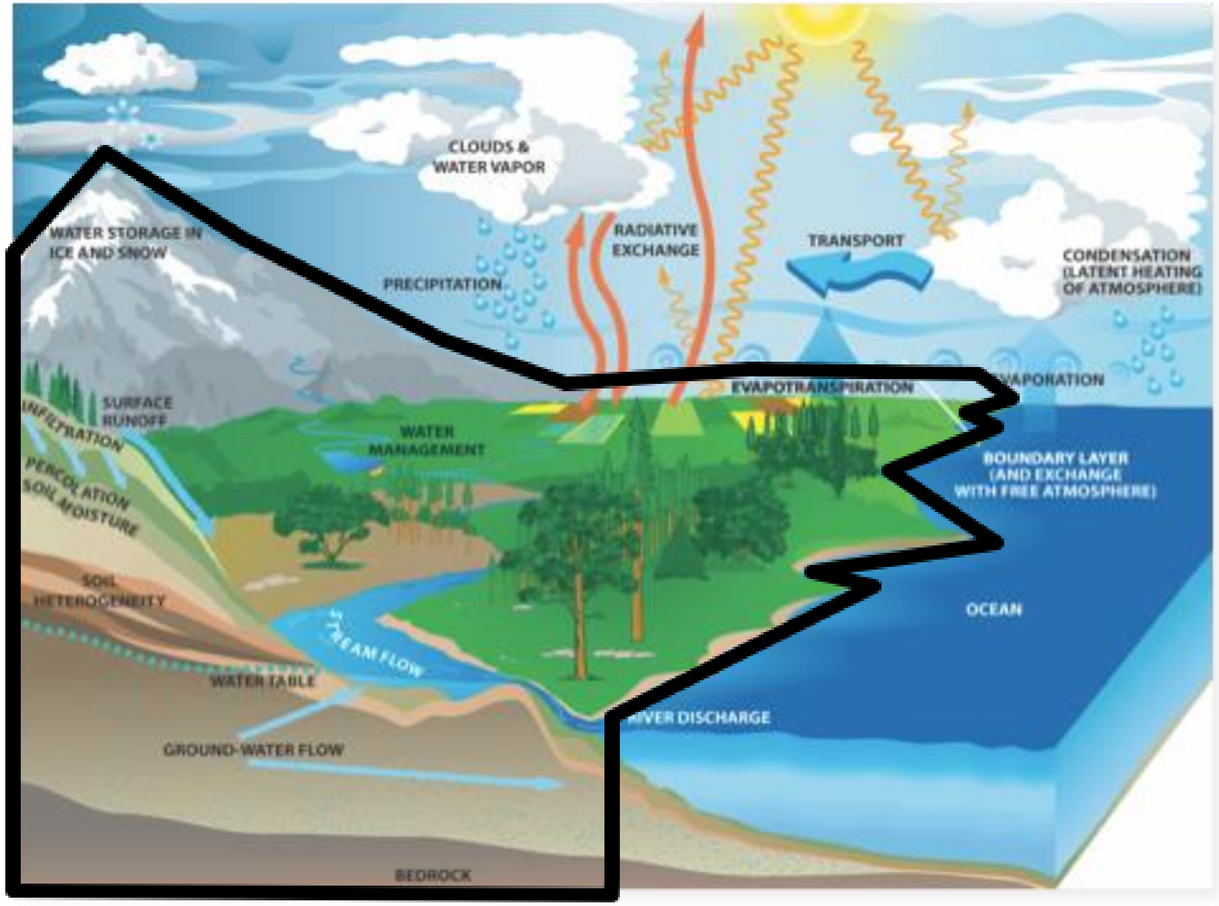
LISF overview

The Land Information System Software Framework

David M. Mocko
SAIC at NASA/GSFC

What is LIS?

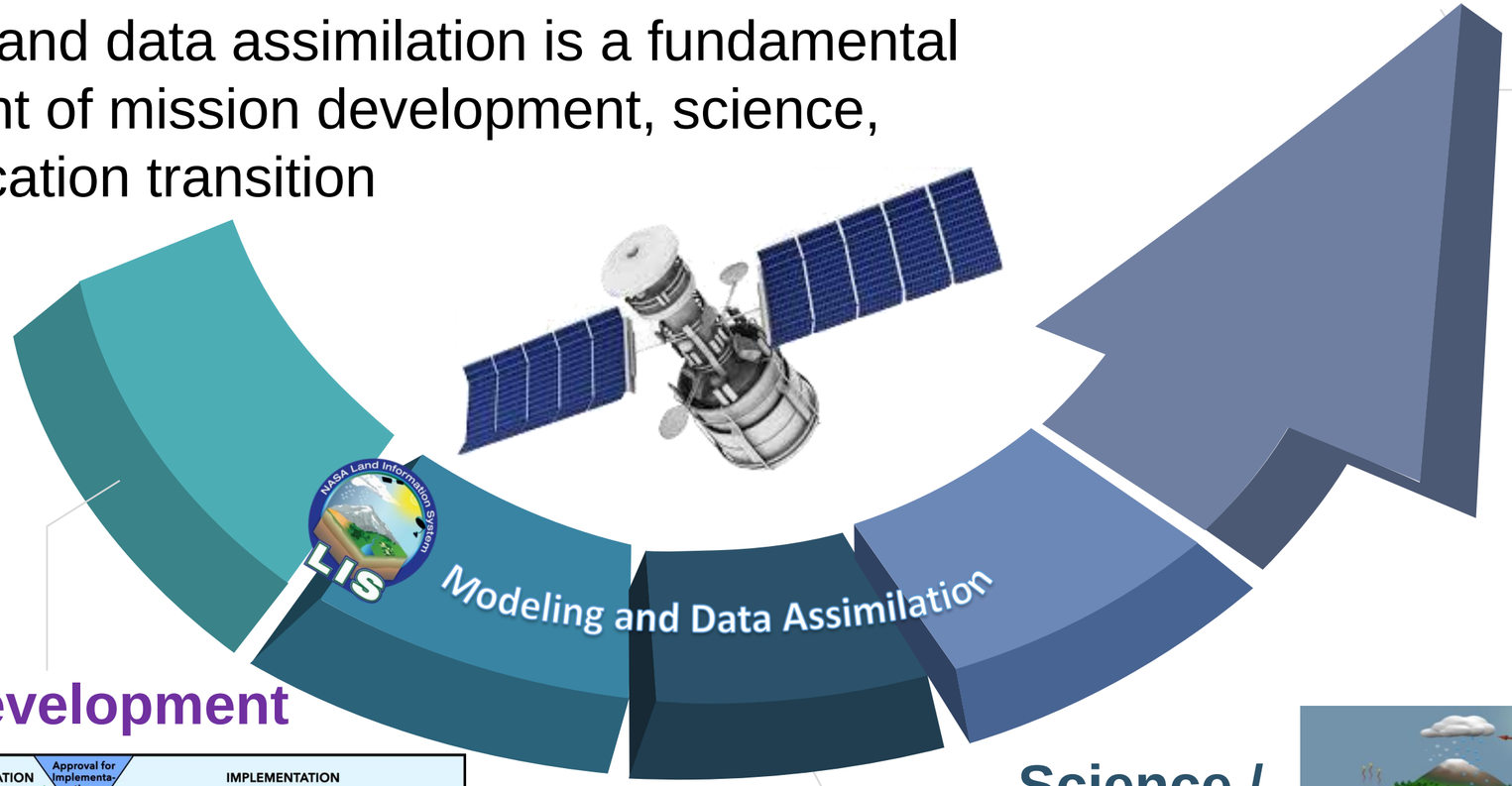
- A system to study land-surface processes and land-atmosphere interactions
- “Use best available observations” to force and constrain the models
- Applications: Weather and climate model initialization, water resources management, natural hazards management, etc.



https://lis.gsfc.nasa.gov/

The Land Information System Software Framework

Modeling and data assimilation is a fundamental component of mission development, science, and application transition



Terrestrial hydrology applications



NWP



Availability



Quality



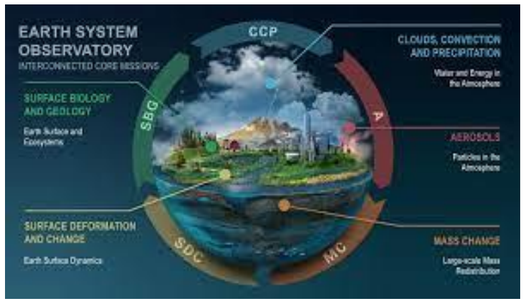
Food security



Extremes

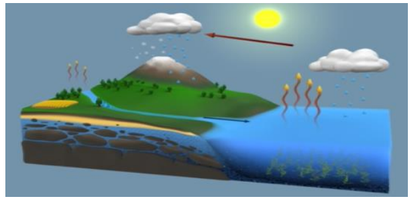
Mission development

LIFECYCLE PHASES	Pre-Phase A: Concept Studies	Approval for Formulation	Phase A: Concept & Technology Development	FORMULATION	Phase B: Preliminary Design & Technology Completion	Approval for Implementation	Phase C: Final Design & Fabrication	IMPLEMENTATION	Phase D: System Assembly, Integration & Test, Launch & Checkout	Phase E: Operations & Sustainment	Phase F: Closeout





Science / research

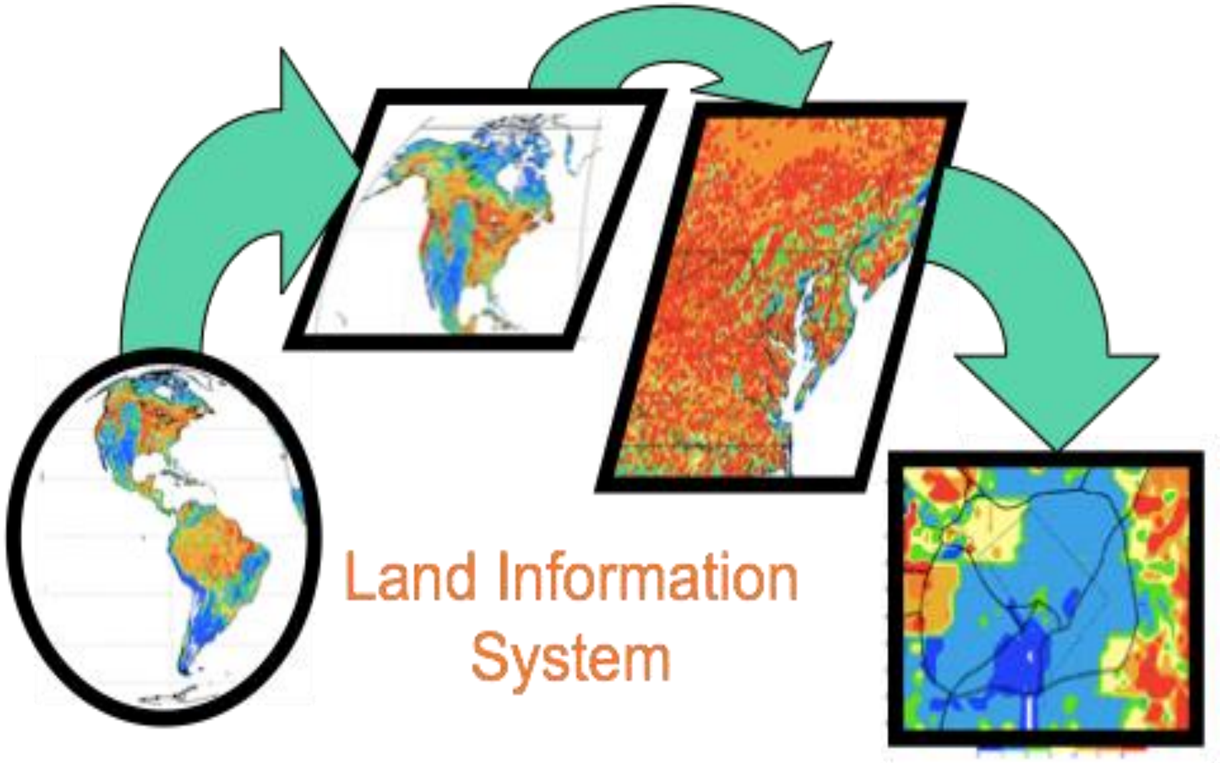


Improved quantification of terrestrial water budget components and their variability

Observation System Simulation Experiments (OSSEs) to quantify the utility of anticipated measurements

Needs fulfilled by LISF

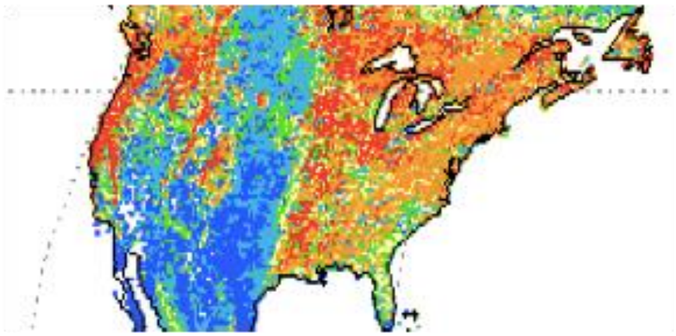
- Need a system viable at different spatial and temporal scales
- Be able to demonstrate the impact of observations at the scale of observations themselves
- Explicit characterization of the land-surface at the same spatial scales as that of cloud and precipitation processes helps in improving the characterization of land-atmosphere interactions



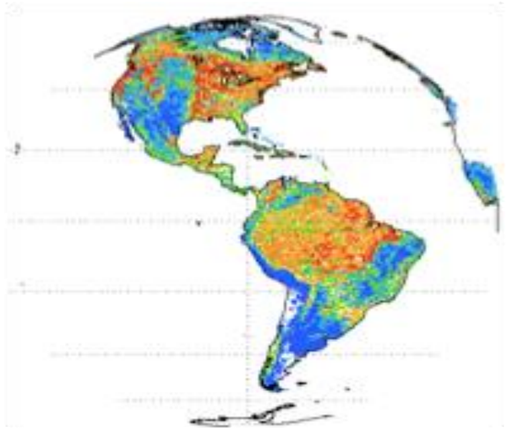
- Need scalable, high-performance computing support to deal with computational challenges
- Need advanced land-surface models and modeling tools (data assimilation, optimization, uncertainty modeling)

LIS - heritage

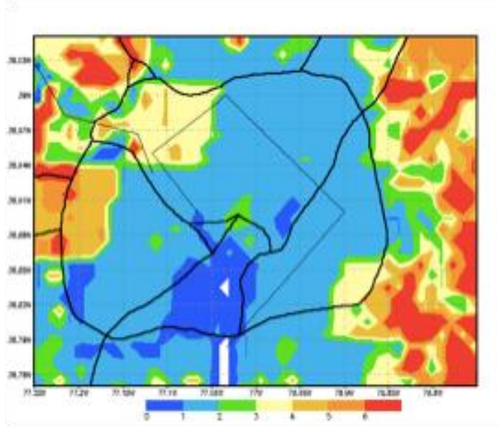
- LIS is capable of modeling at different spatial scales, globally and regionally
- Was developed from a land-surface modeling and data assimilation system (LDAS)
- LIS was a co-winner of NASA's "Software of the Year" award in 2005



North American LDAS
1/8th-degree spatial resolution



Global LDAS
1/4th-degree spatial resolution

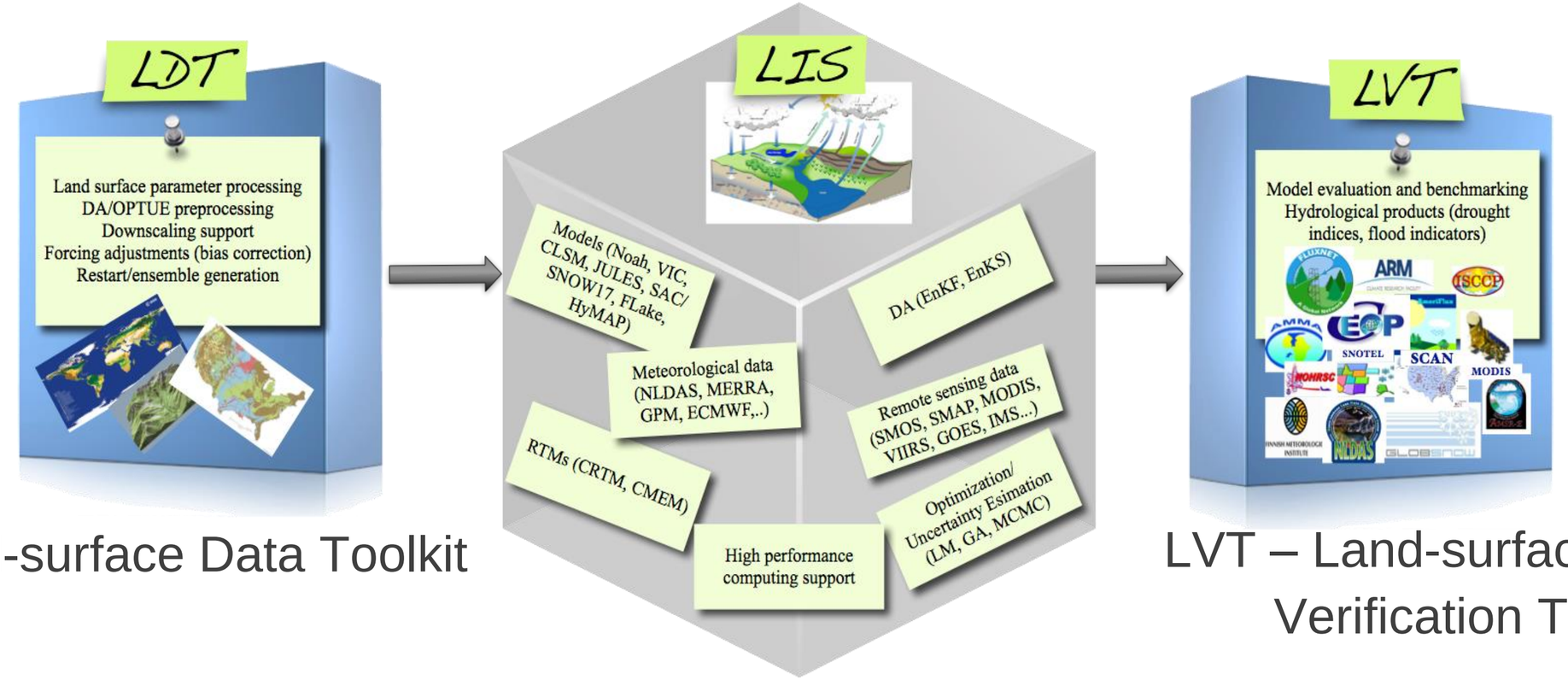


LIS - global, regional, local
up to 1-km and finer

Kumar, S.V., C.D. Peters-Lidard, Y. Tian, P.R. Houser, J. Geiger, S. Olden, L. Lighty, J.L. Eastman, B. Doty, P. Dirmeyer, J. Adams, K. Mitchell, E. F. Wood, and J. Sheffield, 2006: Land Information System - An interoperable framework for high resolution land surface modeling. *Environ. Modeling & Software*, 21, 1402-1415, doi:10.1016/j.envsoft.2005.07.004

Peters-Lidard, C.D., P.R. Houser, Y. Tian, S.V. Kumar, J. Geiger, S. Olden, L. Lighty, B. Doty, P. Dirmeyer, J. Adams, K. Mitchell, E.F. Wood, and J. Sheffield, 2007: High-performance Earth system modeling with NASA/GSFC's Land Information System. *Innovations in Systems and Software Engineering*, 3(3), 157-165, doi:10.1007/s11334-007-0028-x

LISF modeling suite



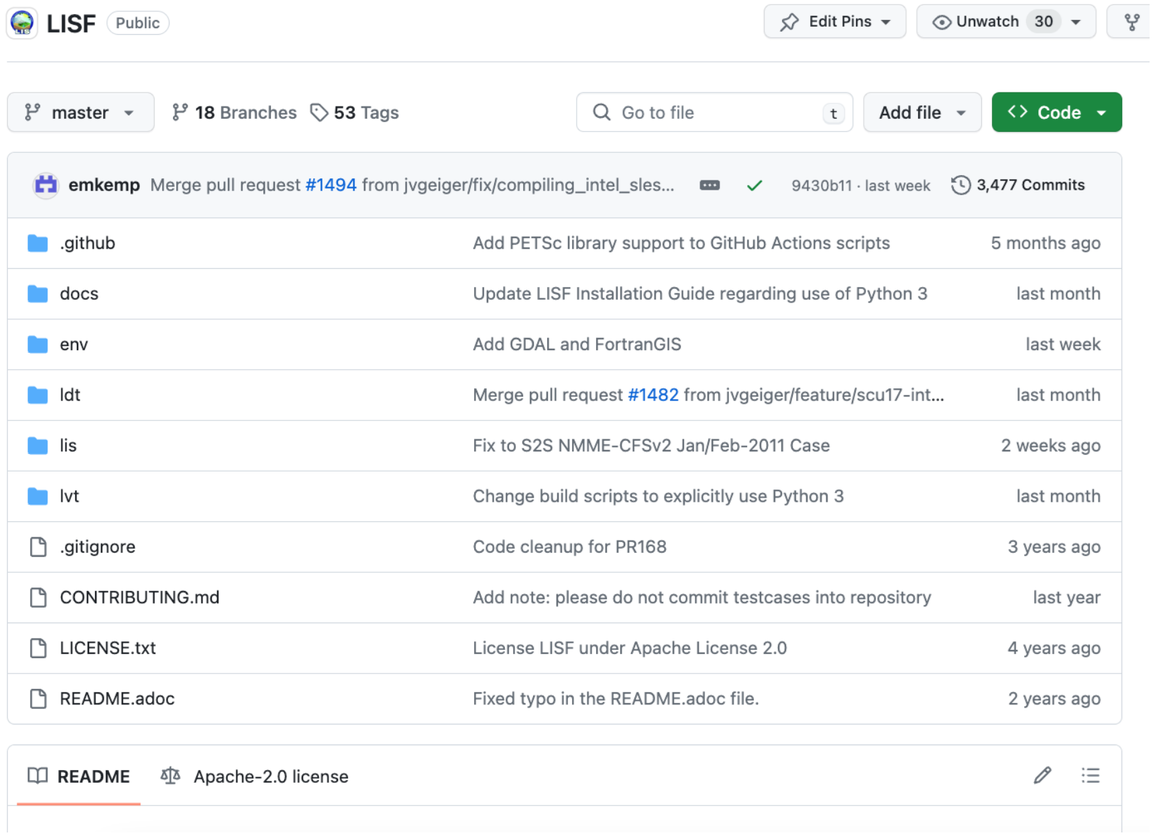
LDT – Land-surface Data Toolkit

LVT – Land-surface Verification Toolkit

- LIS – the modeling and data assimilation framework (with high-performance computing)
- LDT – supports data pre-processing needs (parameter data, data assimilation support, etc.)
- LVT – environment for model benchmarking and evaluation

LISF source code





https://lis.gsfc.nasa.gov/

https://github.com/NASA-LIS/LISF

LISF software requirements

- FORTRAN 90/95 compiler
  - Preferred: Intel, gfortran (GNU), pgi, lahey, absoft
- C compiler
- MPI – if parallel processing capability is desired
- Earth System Modeling Framework (ESMF)
  - 5.x or higher
- All above requirements already installed in container
- LISF supports data in multiple formats (netCDF, HDF, GRIB-1/2, GeoTIFF, etc.)
- LISF is **open-source**, under the Apache-2.0 license



LISF Documentation

LISF Documentation

User Guides

- LISF Installation Guide
- LDT Users' Guide
- LIS Users' Guide
- LVT Users' Guide

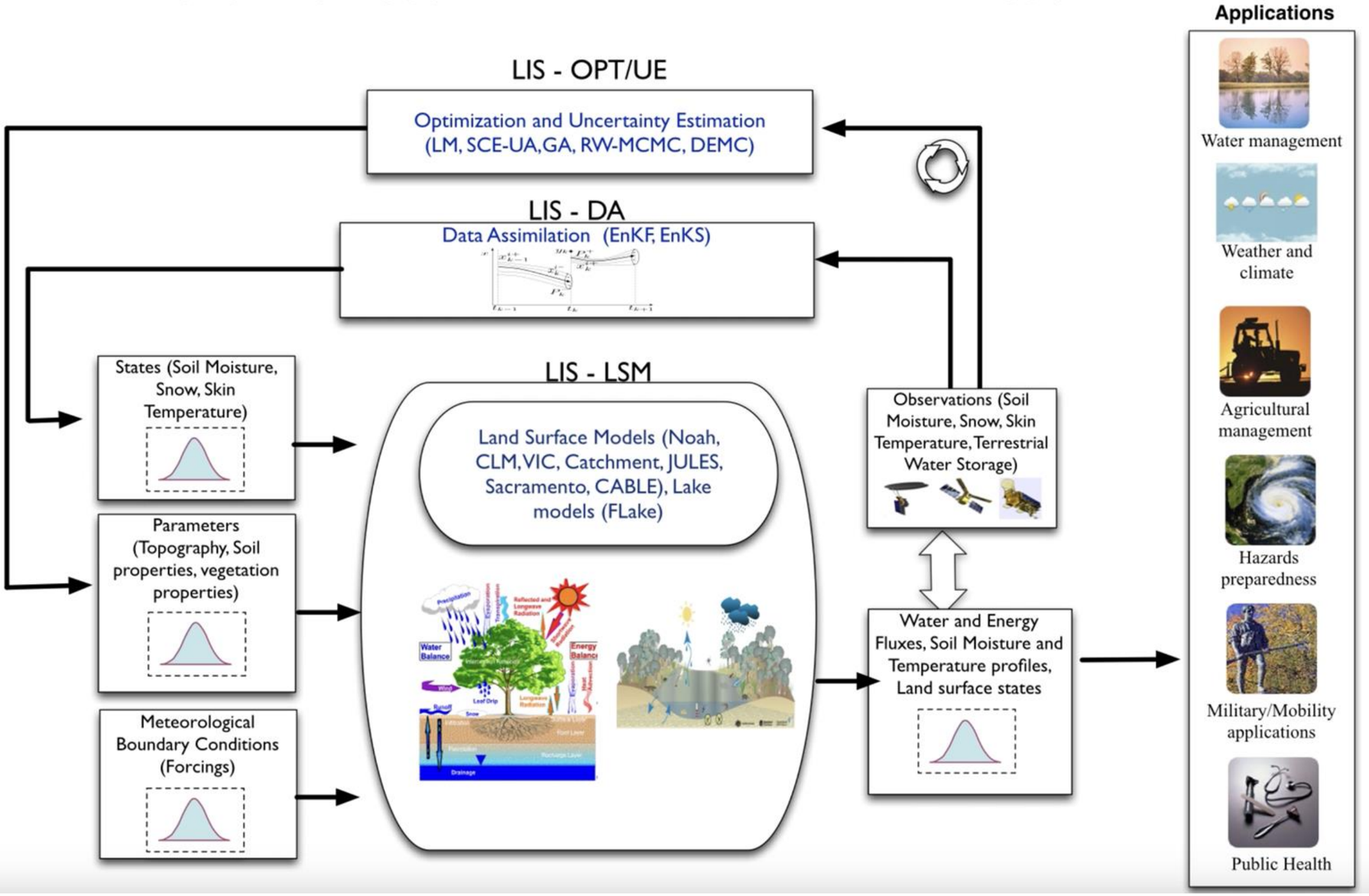
Other Documentation

- Public Testcase Walkthrough
- Quick Start Guide for NCCS Discover Users
- LISF Working with GitHub Guide
- How to Create a LIS Testcase
- Creating a Custom Modulefile

- Installation Guide
- Users' Guides for all 3 components (LDT, LIS, LVT)
- Walkthrough of a public testcase
- Quick Start Guide for NASA users
- How to work with GitHub (for submitting modified LISF code back to the repository)
- How to create a LIS testcase (for new capabilities)
- How to create a custom module file for Linux environments

https://nasa-lis.github.io/LISF/

LIS modes of operation



LIS supports a wide array of land-surface models, meteorological forcing datasets, and remotely-sensed observations for data assimilation.

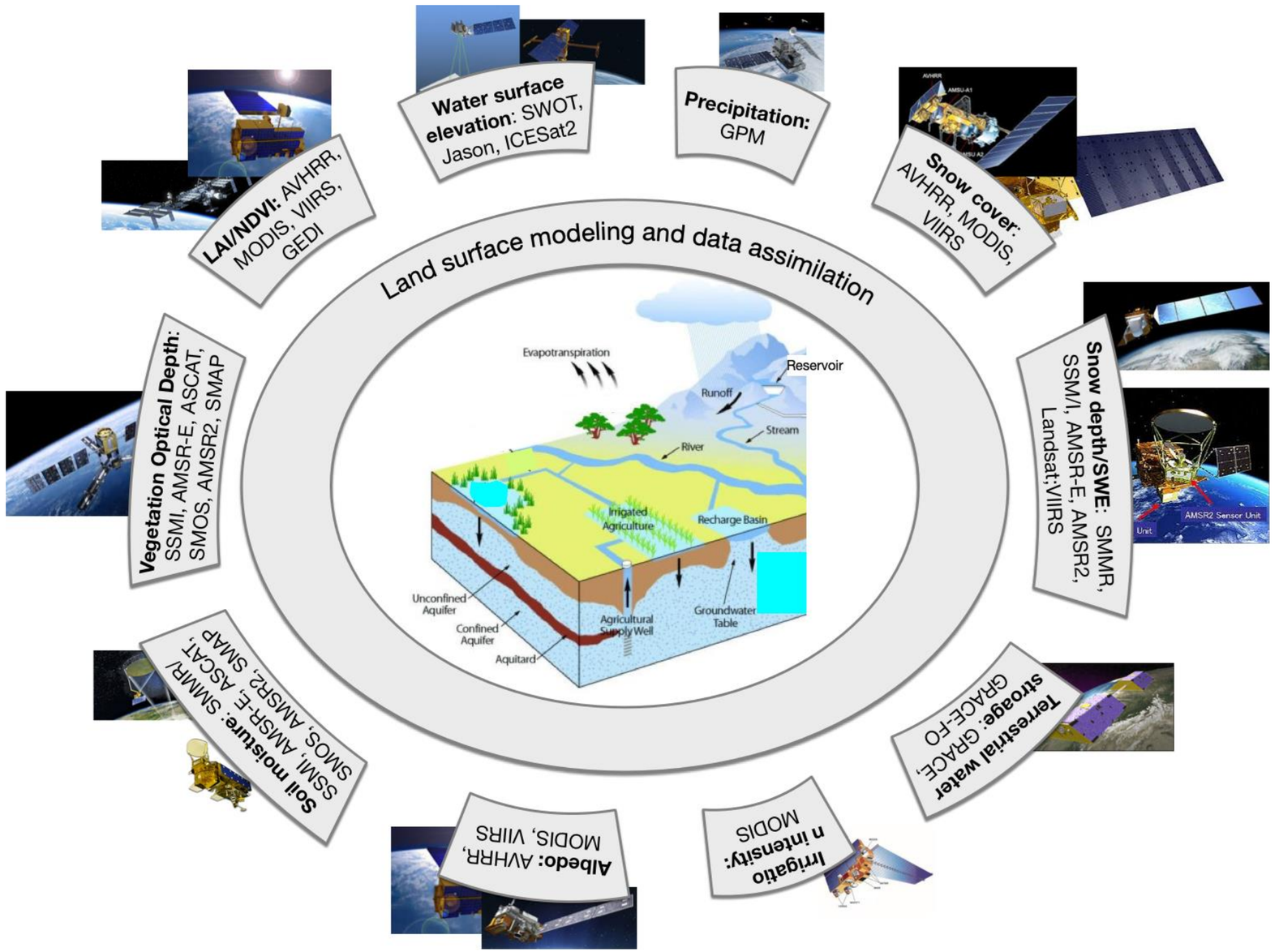
Different data assimilation techniques are supported, as well as parameter optimization and uncertainty estimation (OPT/UE).

All supported via plugins.

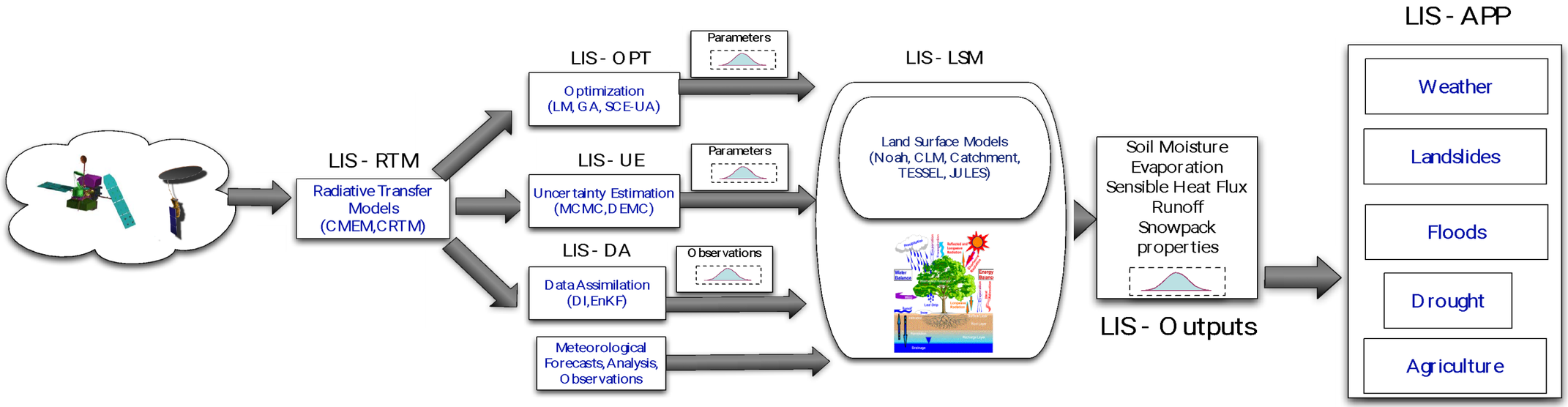
LIS data assimilation support

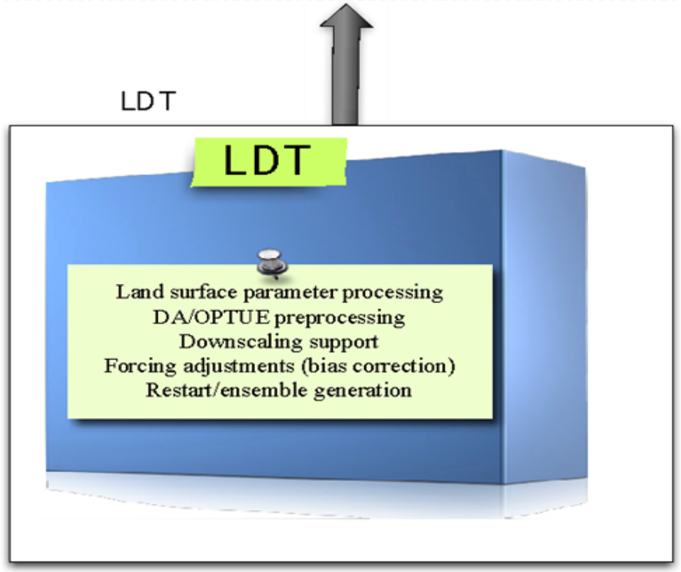
LIS supports data assimilation of numerous remotely-sensed data products that measure:

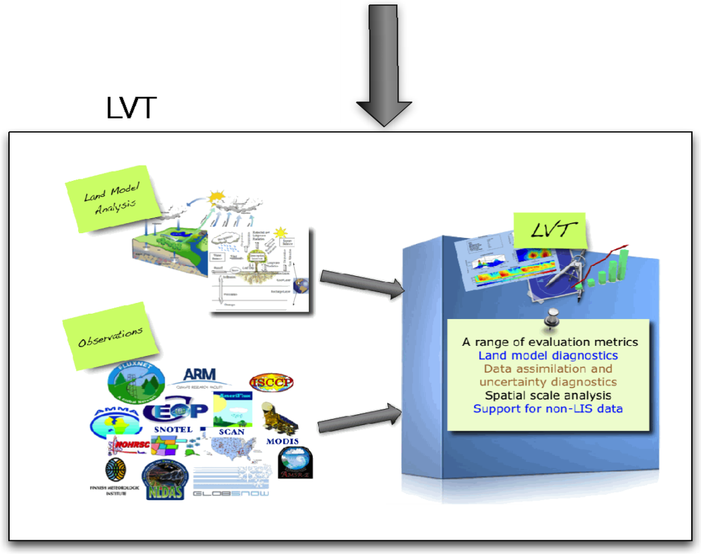
- Precipitation
- Soil moisture
- Snow cover/depth, SWE
- TWS (Terrestrial water storage)
- Irrigation intensity
- Albedo
- Vegetation optical depth
- LAI/NDVI



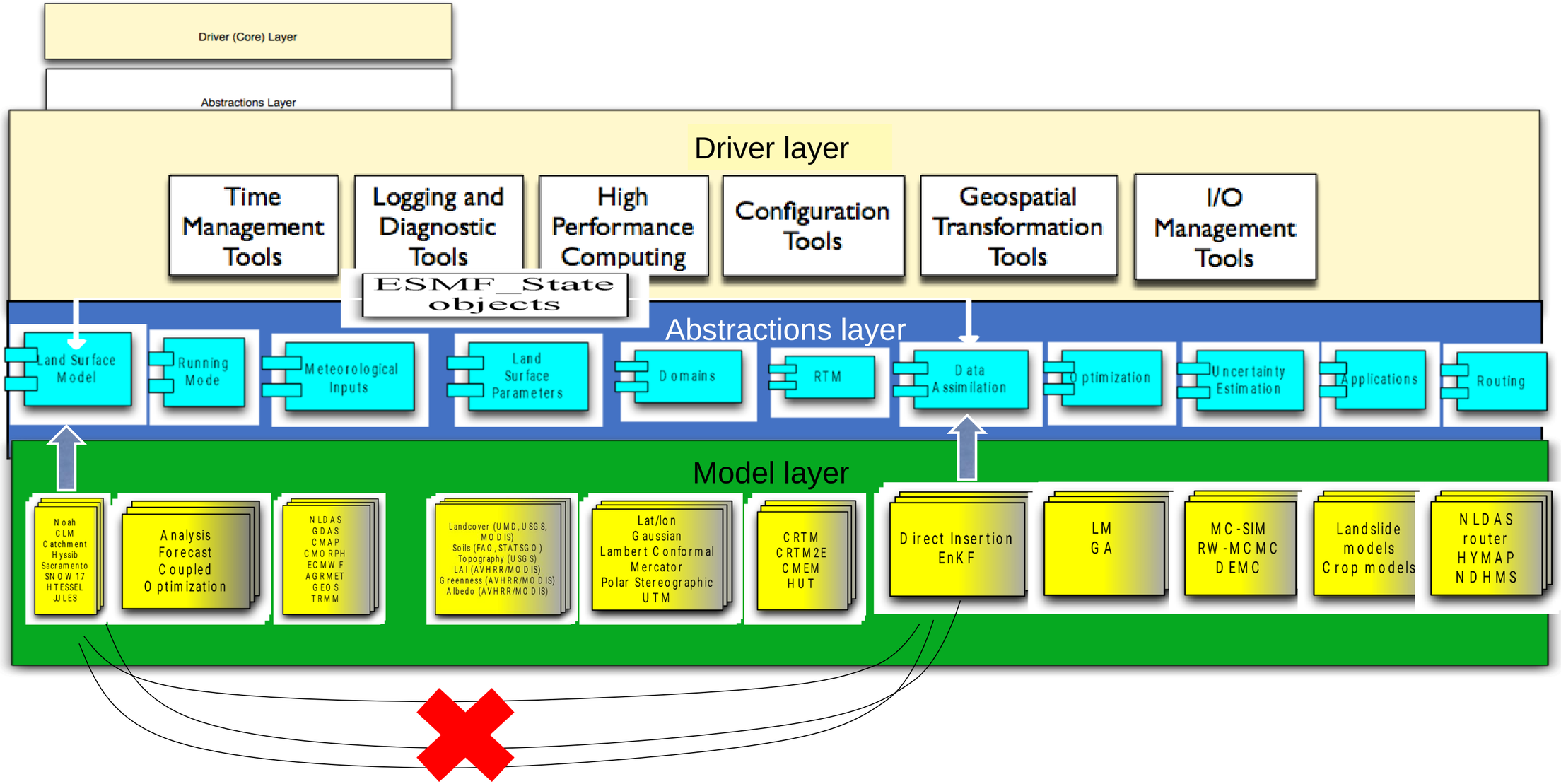
LIS subsystems and toolkits







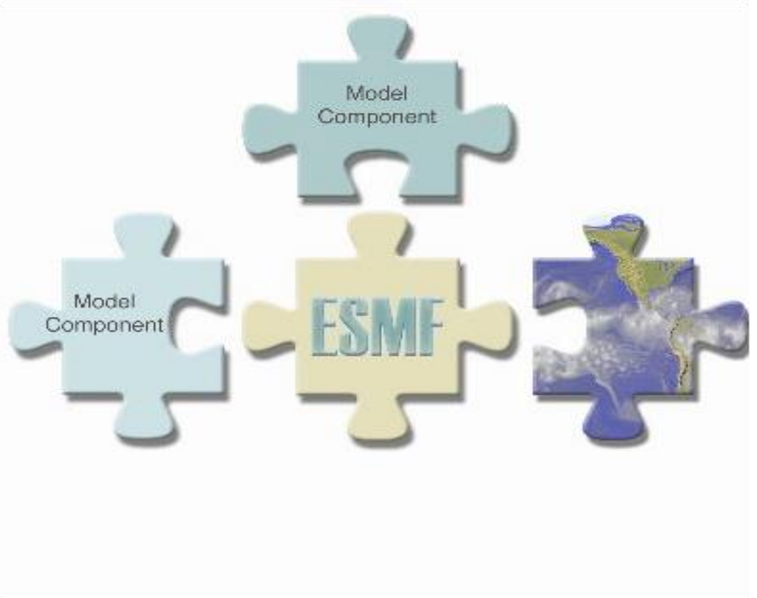
LIS software architecture



What is ESMF?

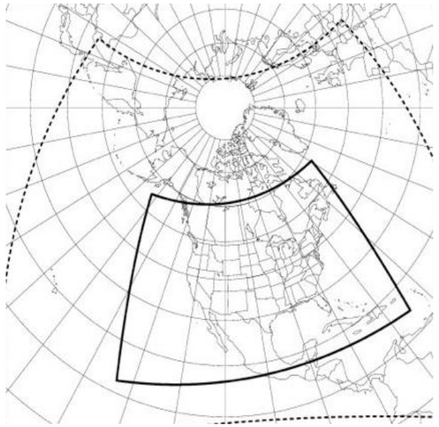
- Software for building and coupling weather, climate and related models
- Provides representations of Earth system grids, tools for mapping between them in multiprocessor environment
- Includes toolkits for building applications: time manager, error handling, resource management, parallel communications
- An application once it is wrapped with ESMF is known as a “Gridded component”
- Gridded components are coupled using “coupler components”

ESMF – Earth System Modeling Framework

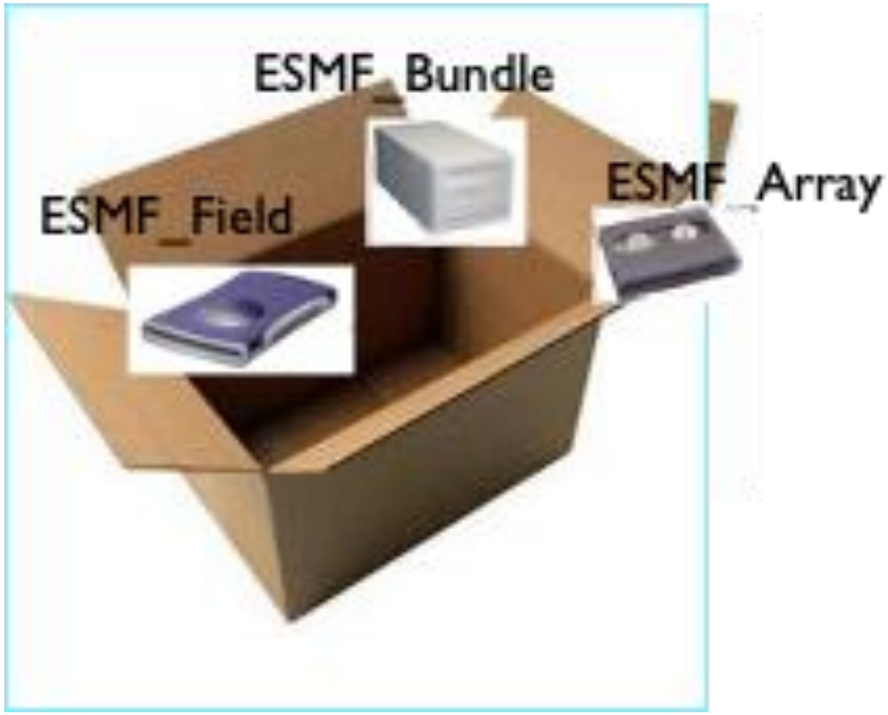


Key ESMF objects

- ESMF_Grid - representation of a grid

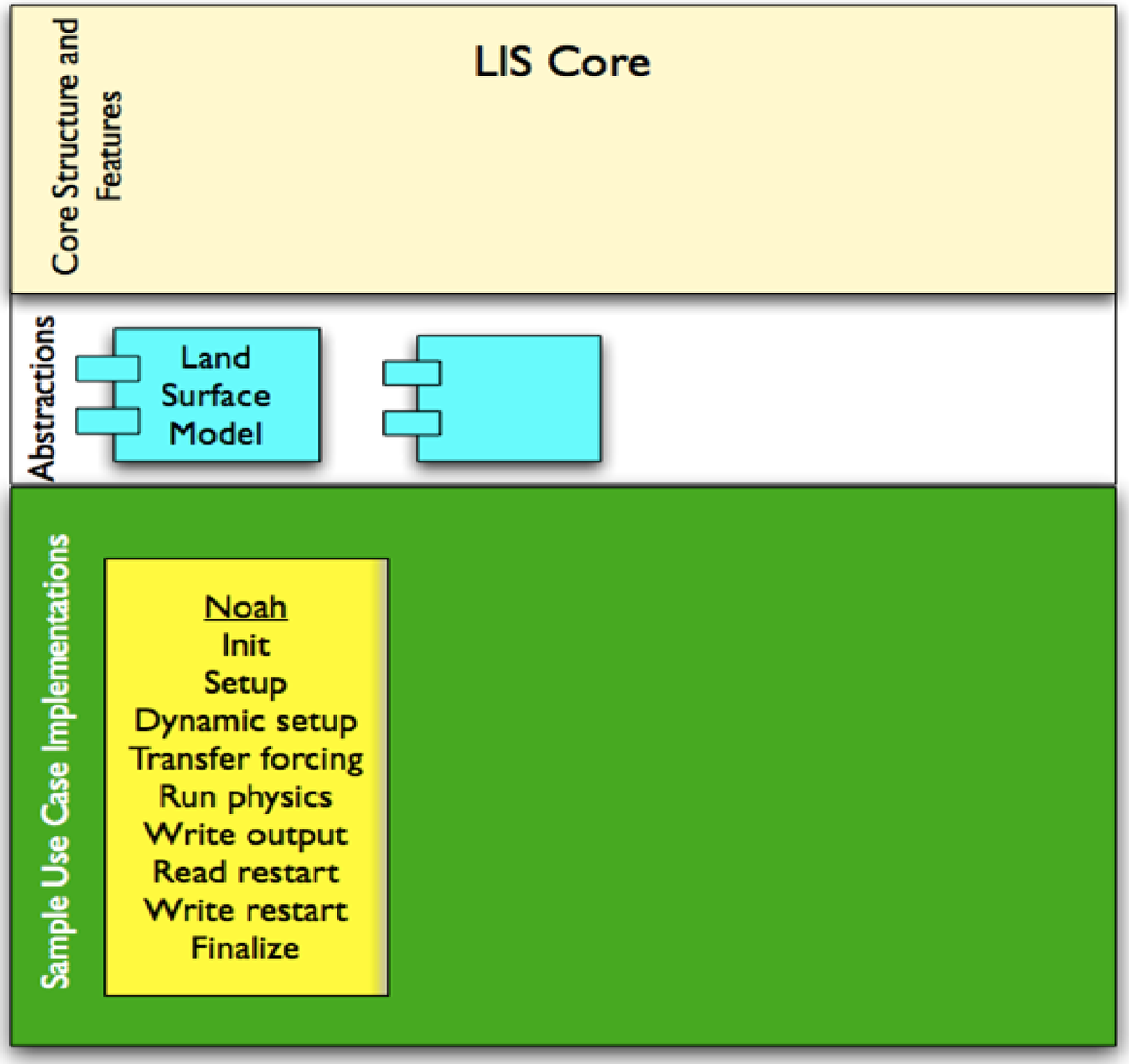


- ESMF_State - objects that hold gridded data
- ESMF_State consists of ESMF_Field, ESMF_Bundle, ESMF_Array



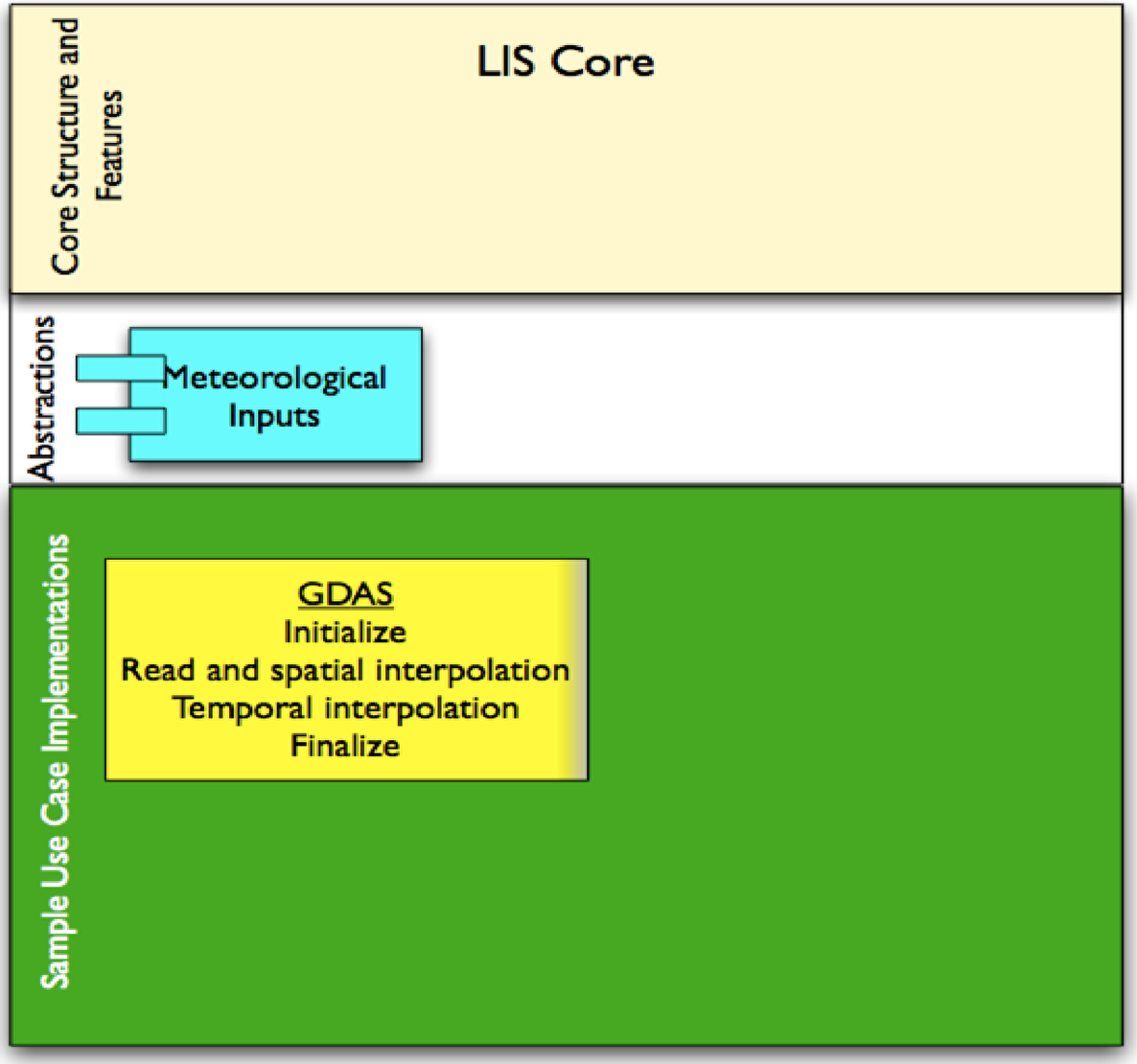
- Data is exchanged between ESMF Gridded components using ESMF_States

Customization: How do we add a new LSM?



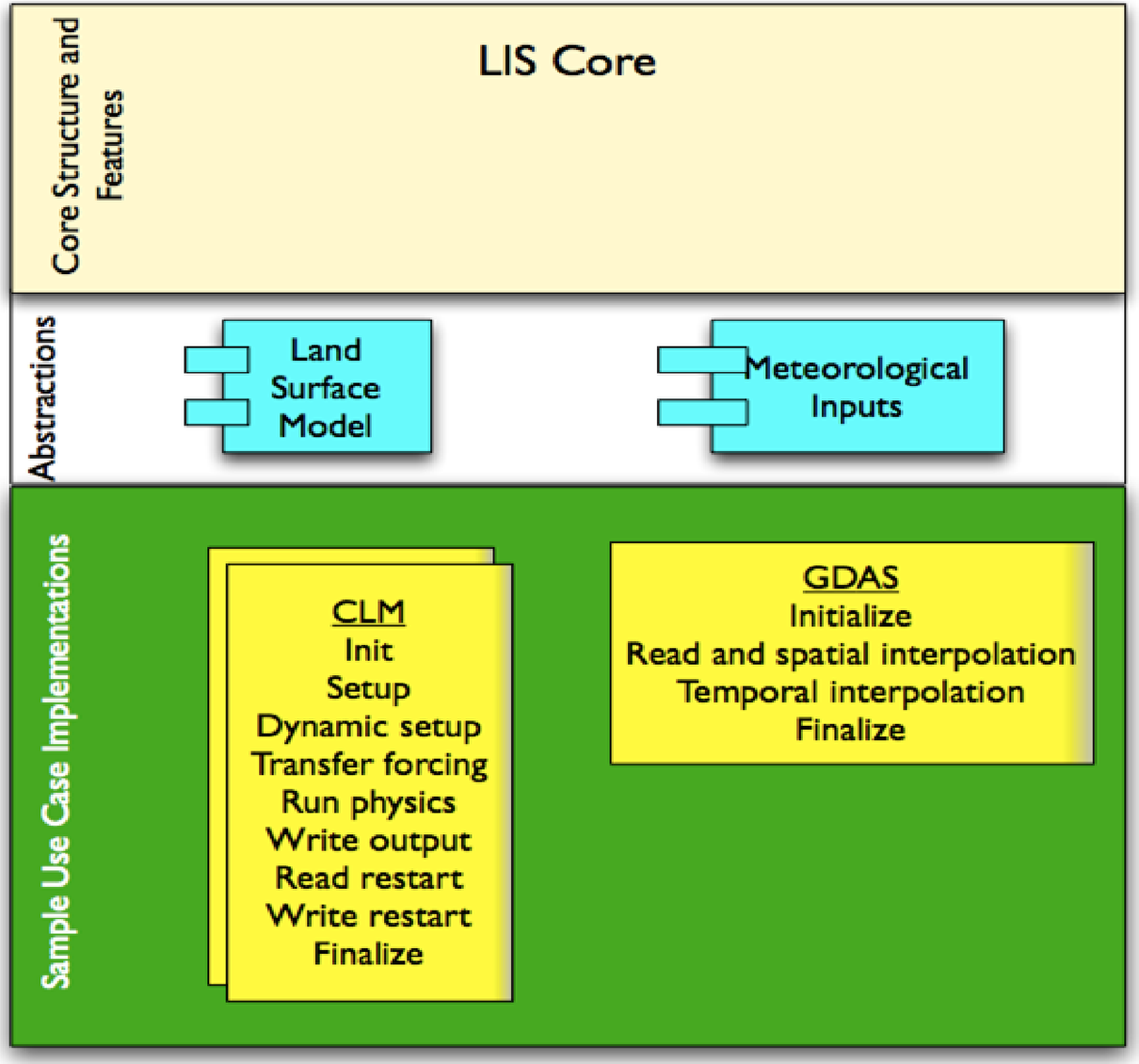
- Need to implement a set of interfaces related to the operation of a land-surface model
- In LIS, these abstract implementations are known as “plugins”
- In the code within the directory: lis/plugins
- Surface models are located in: lis/surfacemodels

Customization: How we add a new met forcing?



- Download (or build) a new meteorological forcing dataset
- Extend the abstract interfaces related to a forcing dataset
- Forcing schemes are located in: lis/metforcing

Combining these components



- LIS provides the “wirings” between the abstract implementations
- Incorporating these components through plugins automatically ensure their integrated and interoperable usage